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United States Patent	12400902
Kind Code	B2
Date of Patent	August 26, 2025
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Carrier ring designs for controlling deposition on wafer bevel/edge

Abstract

Various carrier ring designs and configurations to control an amount of deposition at a wafer's front side and bevel edge are provided. The carrier ring designs can control the amount of deposition at various locations of the wafer while deposition is performed on the wafer's back side, with no deposition desired on the front side of the wafer. These locations include front side, edge, and back side of bevel; and front and back side of the wafer. Edge profiles of the carrier rings are designed to control flow of process gases, flow of front side purge gas, and plasma effects. In some designs, through holes are added to the carrier rings to control gas flows. The edge profiles and added features can reduce or eliminate deposition at the wafer's front side and bevel edge.

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Appl. No.: 18/494756

Filed: October 25, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240055293 A1	Feb. 15, 2024

Related U.S. Application Data

continuation parent-doc WO PCT/US2021/015487 20210128 PENDING child-doc US 17644757
division parent-doc US 17644757 20211216 US 11830759 child-doc US 18494756
us-provisional-application US 62975146 20200211

Publication Classification

Int. Cl.: H01L21/687 (20060101); H01J37/32 (20060101)

U.S. Cl.:

CPC H01L21/68721 (20130101); H01J37/32642 (20130101); H01J37/32715 (20130101); H01L21/68735 (20130101); H01L21/68785 (20130101); H01J2237/332 (20130101)

Field of Classification Search

CPC: H01L (21/68721); H01L (21/68735); H01L (21/68785); H01J (37/32642); H01J (37/32715); H01J (2237/332)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5304248	12/1993	Cheng et al.	N/A	N/A
5476548	12/1994	Lei et al.	N/A	N/A
5552124	12/1995	Su	N/A	N/A
5769951	12/1997	van de Ven et al.	N/A	N/A
5810931	12/1997	Stevens et al.	N/A	N/A
5848889	12/1997	Tietz et al.	N/A	N/A
6040011	12/1999	Yudovsky et al.	N/A	N/A
6048403	12/1999	Deaton et al.	N/A	N/A
6096135	12/1999	Guo et al.	N/A	N/A
6126382	12/1999	Scales et al.	N/A	N/A
6143082	12/1999	McInerney et al.	N/A	N/A
6159299	12/1999	Koai et al.	N/A	N/A
6280183	12/2000	Mayur et al.	N/A	N/A
6296712	12/2000	Guo et al.	N/A	N/A
6343784	12/2001	Jourde	451/364	H01L 21/68785
6350320	12/2001	Sherstinsky et al.	N/A	N/A
6374512	12/2001	Guo et al.	N/A	N/A
9738975	12/2016	Gomm et al.	N/A	N/A
9870917	12/2017	Kang	N/A	H01L 21/02274
10062598	12/2017	Ngo et al.	N/A	N/A
10242848	12/2018	Jeon et al.	N/A	N/A
11018045	12/2020	Seo	N/A	C23C 14/50
11098406	12/2020	Jeon	N/A	C23C 16/50
11830759	12/2022	Janicki et al.	N/A	N/A
11837495	12/2022	Janicki et al.	N/A	N/A
2001/0042514	12/2000	Mizuno et al.	N/A	N/A
2004/0020599	12/2003	Tanaka et al.	N/A	N/A
2005/0196971	12/2004	Sen et al.	N/A	N/A
2005/0245005	12/2004	Benson	N/A	N/A

2007/0173059	12/2006	Young et al.	N/A	N/A
2007/0235660	12/2006	Hudson	N/A	N/A
2008/0289766	12/2007	Heemstra et al.	N/A	N/A
2011/0263123	12/2010	Gomi et al.	N/A	N/A
2012/0018940	12/2011	Kumnig et al.	N/A	N/A
2012/0213500	12/2011	Koelmel et al.	N/A	N/A
2013/0000558	12/2012	Hara et al.	N/A	N/A
2013/0269737	12/2012	Mizuno	N/A	N/A
2014/0113458	12/2013	Pan	438/795	H01L 21/68735
2014/0235063	12/2013	McMillin et al.	N/A	N/A
2014/0261187	12/2013	Krishnan et al.	N/A	N/A
2014/0273460	12/2013	Reyland	438/694	H01L 21/76805
2014/0335698	12/2013	Singh et al.	N/A	N/A
2015/0020736	12/2014	Pan et al.	N/A	N/A
2015/0047564	12/2014	Kim	N/A	N/A
2015/0187620	12/2014	Gurary et al.	N/A	N/A
2015/0221539	12/2014	Ouye	N/A	N/A
2016/0002778	12/2015	Ravi et al.	N/A	N/A
2016/0024650	12/2015	Toyoda et al.	N/A	N/A
2016/0177444	12/2015	Baldasseroni et al.	N/A	N/A
2016/0289827	12/2015	Augustyniak	N/A	C23C 16/52
2016/0307742	12/2015	Mishra et al.	N/A	N/A
2016/0355928	12/2015	Toriya et al.	N/A	N/A
2017/0002465	12/2016	Shaikh et al.	N/A	N/A
2017/0278681	12/2016	Qi et al.	N/A	N/A
2018/0090344	12/2017	Kim et al.	N/A	N/A
2018/0112309	12/2017	Kamio et al.	N/A	N/A
2018/0122685	12/2017	Breiling	N/A	C23C 16/5096
2018/0138074	12/2017	Lee et al.	N/A	N/A
2019/0062918	12/2018	Shaikh et al.	N/A	N/A
2019/0153592	12/2018	Bois et al.	N/A	N/A
2020/0202291	12/2019	Yamamoto et al.	N/A	N/A
2020/0234996	12/2019	Rokkam	N/A	H01L 21/6875
2020/0248307	12/2019	Gurary et al.	N/A	N/A
2021/0062325	12/2020	Ramalingam et al.	N/A	N/A
2021/0141312	12/2020	Nakiboglu	N/A	H01L 21/6875
2021/0202217	12/2020	Lim et al.	N/A	N/A
2021/0375591	12/2020	Chandrashekar et al.	N/A	N/A
2022/0108912	12/2021	Janicki et al.	N/A	N/A
2022/0115261	12/2021	Janicki et al.	N/A	N/A
2022/0259731	12/2021	Tsuji	N/A	N/A
2022/0293398	12/2021	Jeong et al.	N/A	N/A
2023/0040885	12/2022	Gulabal et al.	N/A	N/A

2023/0238223	12/2022	Linebarger, Jr. et al.	N/A	N/A
2025/0014938	12/2024	Chandrashekar et al.	N/A	N/A
2025/0087465	12/2024	Ko	N/A	H01L 21/67248

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1930322	12/2006	CN	N/A
201075384	12/2007	CN	N/A
S59132623	12/1983	JP	N/A
H04268724	12/1991	JP	N/A
H0537960	12/1992	JP	N/A
H0881775	12/1995	JP	N/A
H0917747	12/1996	JP	N/A
H0982653	12/1996	JP	N/A
H10298767	12/1997	JP	N/A
H11219918	12/1998	JP	N/A
2001329370	12/2000	JP	N/A
2001527285	12/2000	JP	N/A
2002110652	12/2001	JP	N/A
2002518601	12/2001	JP	N/A
2002190466	12/2001	JP	N/A
2002246370	12/2001	JP	N/A
2002363756	12/2001	JP	N/A
2003027242	12/2002	JP	N/A
2003524703	12/2002	JP	N/A
2006502016	12/2005	JP	N/A
2008072122	12/2007	JP	N/A
2010045200	12/2009	JP	N/A
2011035026	12/2010	JP	N/A
2015536048	12/2014	JP	N/A
2016530705	12/2015	JP	N/A
970052044	12/1996	KR	N/A
20040094240	12/2003	KR	N/A
20060036846	12/2005	KR	N/A
20070031785	12/2006	KR	N/A
20110135631	12/2010	KR	N/A
20140045806	12/2013	KR	N/A
20150138675	12/2014	KR	N/A
20160029073	12/2015	KR	N/A
20160075351	12/2015	KR	N/A
20160117261	12/2015	KR	N/A
20180053258	12/2017	KR	N/A
101937692	12/2018	KR	N/A
20190051291	12/2018	KR	N/A
20190056021	12/2018	KR	N/A
20200000460	12/2019	KR	N/A

20200038317	12/2019	KR	N/A
20200135554	12/2019	KR	N/A
452838	12/2000	TW	N/A
201401428	12/2013	TW	N/A
D160525	12/2013	TW	N/A
201546937	12/2014	TW	N/A
201701318	12/2016	TW	N/A
201742948	12/2016	TW	N/A
D207742	12/2019	TW	N/A
WO-2014209489	12/2013	WO	N/A
WO-2019204754	12/2018	WO	N/A
WO-2019204754	12/2019	WO	N/A
WO-2021146352	12/2020	WO	N/A
WO-2021179000	12/2020	WO	N/A

OTHER PUBLICATIONS

KR Office Action dated Jul. 26, 2022, in Application No. KR10-2022-7024631. cited by applicant
KR Search Report dated Jul. 25, 2022, in Application No. KR10-2022-7024631, with English translation. cited by applicant

JP Office Action dated Oct. 8, 2024 in JP Application No. 2022-547668 with English translation. cited by applicant

KR Office Action dated Sep. 26, 2024 in KR Application No. 10-2020-7033442 with English Translation. cited by applicant

TW Office Action dated Sep. 2, 2024 in TW Application No. 110101601 with English translation. cited by applicant

CN Office Action dated Jan. 8, 2024 in CN Application No. 202230757795.6 with English Translation. cited by applicant

CN Office Action dated Jul. 6, 2023, in application No. CN202230757795.6 with English translation. cited by applicant

CN Office Action dated May 24, 2024 in CN Application No. 201980036351.9 with English translation. cited by applicant

CN Office Action dated May 31, 2024 in CN Application No. 202230757795.6, with English Translation. cited by applicant

Inlet. "Merriam-Webster.com Dictionary, Merriam-Webster," [<https://www.merriam-webster.com/dictionary/inlet>]. Accessed Jun. 21, 2024. cited by applicant

International Preliminary Report on Patentability and written opinion dated Aug. 25, 2022 in Application PCT/US2021/015487. cited by applicant

International Search Report and Written Opinion dated Oct. 13, 2021, in application No. PCT/US2021/038210. cited by applicant

International Preliminary Report on Patentability and Written Opinion dated Jun. 6, 2024 in PCT Application No. PCT/US2022/080276. cited by applicant

International Preliminary Report on Patentability dated Oct. 29, 2020 in PCT Application PCT/US2019/028362. cited by applicant

International Preliminary Report on Patentability dated Jan. 5, 2023 in PCT Application No. PCT/US2021/038210. cited by applicant

International Preliminary Report on Patentability dated Jul. 28, 2022 in PCT Application No. PCT/US2021/013327. cited by applicant

International Search Report & Written Opinion dated Aug. 9, 2019 in PCT Application PCT/US2019/028362. cited by applicant

International Search Report and Written Opinion dated Apr. 4, 2023 in Application No.

PCT/US2022/080276. cited by applicant
International Search Report and Written Opinion dated Apr. 29, 2021 in PCT Application No. PCT/US2021/013327. cited by applicant
International Search Report and Written Opinion dated May 20, 2021 in PCT Application No. PCT/US2021/015487. cited by applicant
International Search Report and Written Opinion dated Sep. 4, 2023 in PCT Application No. PCT/US2023/021778. cited by applicant
JP Office Action dated Mar. 10, 2023 in Application No. JP2022-24512 with English Translation. cited by applicant
JP Office Action dated May 30, 2023 in Application No. JP 2020-557974 with English translation. cited by applicant
KR Office Action dated Apr. 3, 2022, in Application No. KR10-2022-7001497 with English translation. cited by applicant
KR Office Action dated Apr. 3, 2022, in Application No. KR1020227001591 with English translation. cited by applicant
KR Office Action dated Feb. 15, 2024 in KR Application No. 10-2023-7025656 with English translation. cited by applicant
KR Office Action dated Feb. 21, 2023, in Application No. KR10-2022-7024631 with English translation. cited by applicant
KR Office Action dated Jan. 22, 2024 in KR Application No. 10-2020-7033442 with English Translation. cited by applicant
KR Office Action dated Jul. 26, 2022, in Application No. KR10-2022-7024631 with English translation. cited by applicant
KR Office Action dated Jul. 27, 2023, in Application No. KR10-2022-7024631 with English Translation. cited by applicant
KR Office Action dated Jun. 21, 2023, in Application No. KR30-2022-0046450 with English translation. cited by applicant
KR Office Action dated Jun. 21, 2023, in Application No. KR30-2022-0046451 with English translation. cited by applicant
KR Office Action dated Mar. 18, 2016 in KR Application No. 10-2012-0004935. cited by applicant
KR Office Action dated Mar. 28, 2023, in Application No. KR 10-2022-7001591 with English translation. cited by applicant
KR Office Action dated Mar. 28, 2023, in Application No. KR10-2022-7001497 with English translation. cited by applicant
KR Office Action dated Nov. 28, 2024 in KR Application No. 10-2023-7025656, with English Translation. cited by applicant
KR Office Action dated Oct. 31, 2023, in Application No. KR10-2022-7001591 with English Translation. cited by applicant
KR Office Action dated Sep. 2, 2022, in Application No. KR10-2022-7024631 with English translation. cited by applicant
KR Office Action dated Sep. 26, 2022 in Application No. KR10-2022-7001497 with English translation. cited by applicant
KR Office Action dated Sep. 26, 2022 in Application No. KR10-2022-7001591 with English translation. cited by applicant
Over. "Merriam-Webster.com Dictionary, Merriam-Webster," [<https://www.merriam-webster.com/dictionary/over>]. Accessed Jun. 21, 2024. cited by applicant
TW Office Action dated Apr. 15, 2024 in TW Application No. 111305560, with English Translation. cited by applicant
TW Office Action dated Dec. 7, 2023, in TW application No. 111305560 with English translation. cited by applicant

TW Office Action dated Dec. 13, 2022 in Application No. TW108113695 with English translation. cited by applicant

TW Office Action dated Dec. 15, 2023 in TW Application No. 111305554 with English Translation. cited by applicant

TW Office Action dated Jul. 26, 2024 in TW Application No. 111305554 with English translation. cited by applicant

TW Office Action dated Jul. 31, 2023 in Application No. TW111305554 with English Translation. cited by applicant

TW Office Action dated May 23, 2024 in TW Application No. 110104695 with English translation. cited by applicant

TW Search Report & Grant Decision dated Jan. 25, 2015 in TW Application No. 102116792. cited by applicant

U.S. Final office Action dated Jun. 28, 2022 in U.S. Appl. No. 17/644,754. cited by applicant

U.S. Advisory Action dated Jun. 27, 2023, in U.S. Appl. No. 17/644,757. cited by applicant

U.S. Advisory Action dated Nov. 1, 2022, in U.S. Appl. No. 17/644,757. cited by applicant

U.S. Corrected Notice of Allowance dated Nov. 3, 2022 in U.S. Appl. No. 17/644,754. cited by applicant

U.S. Corrected Notice of Allowance dated Sep. 28, 2022 in U.S. Appl. No. 17/644,754. cited by applicant

U.S. Final Office Action dated Apr. 5, 2023 in U.S. Appl. No. 17/644,757. cited by applicant

U.S. Final Office Action dated Jul. 27, 2022, in U.S. Appl. No. 17/644,757. cited by applicant

U.S. Final Office Action dated Jun. 27, 2024 in U.S. Appl. No. 15/733,766. cited by applicant

U.S. Non-Final office Action dated Jan. 12, 2023 in U.S. Appl. No. 17/644,754. cited by applicant

U.S. Non-Final Office Action dated Dec. 14, 2022 in U.S. Appl. No. 17/644,757. cited by applicant

U.S. Non-Final Office Action dated Nov. 21, 2023 in U.S. Appl. No. 15/733,766. cited by applicant

U.S. Notice of Allowance dated Aug. 10, 2023 in U.S. Appl. No. 17/644,754. cited by applicant

U.S. Notice of Allowance dated Jul. 20, 2023, in U.S. Appl. No. 17/644,757. cited by applicant

U.S. Notice of Allowance dated Jul. 31, 2023 in U.S. Appl. No. 17/644,754. cited by applicant

U.S. Notice of Allowance dated May 5, 2023 in U.S. Appl. No. 17/644,754. cited by applicant

U.S. Notice of Allowance dated Nov. 3, 2023 in U.S. Appl. No. 17/644,754. cited by applicant

U.S. Notice of Allowance dated Sep. 12, 2023 in U.S. Appl. No. 17/644,757. cited by applicant

U.S. Notice of Allowance dated Sep. 16, 2022 in U.S. Appl. No. 17/644,754. cited by applicant

US Office Action dated Feb. 15, 2022 in U.S. Appl. No. 17/644,754. cited by applicant

US Office Action dated Mar. 17, 2022 in U.S. Appl. No. 17/644,757. cited by applicant

U.S. Appl. No. 17/758,797, inventors Gulabal et al., filed on Jul. 13, 2022. cited by applicant

U.S. Appl. No. 18/712,211, inventor Chandrashekar A, filed on May 21, 2024. cited by applicant

U.S. Appl. No. 29/864,203, inventors Huang et al., filed on May 13, 2022. cited by applicant

U.S Restriction requirement dated Aug. 7, 2023 in U.S. Appl. No. 15/733,766. cited by applicant

Decision from the Korean Patent Court in KR application 10-2022-7001591, decided Jan. 24, 2025, dismissing Appeal against the Decision of the IPTAB upholding Final Rejection by the KIPO (translation appended). cited by applicant

JP Office Action dated Apr. 8, 2025 in JP Application No. 2022-547668, with English Translation. cited by applicant

SG Search Report and Written Opinion dated Jan. 29, 2025 in SG Application No. 11202251767J. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) An Application Data Sheet is filed concurrently with this specification as part of the present application. Each application that the present application claims benefit of or priority to as identified in the concurrently filed Application Data Sheet is incorporated by reference herein in its entirety and for all purposes.

FIELD

(2) The present disclosure relates generally to substrate processing systems and more particularly to carrier ring designs for controlling deposition on wafer bevel/edge.

BACKGROUND

(3) The background description provided here is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

(4) A substrate processing system typically includes a plurality of processing chambers (also called process modules) to perform deposition, etching, and other treatments on substrates such as semiconductor wafers. Examples of processes that may be performed on a substrate include, but are not limited to, a plasma enhanced chemical vapor deposition (PECVD) process, a chemically enhanced plasma vapor deposition (CEPVD) process, and a sputtering physical vapor deposition (PVD) process. Additional examples of processes that may be performed on a substrate include, but are not limited to, etching (e.g., chemical etching, plasma etching, reactive ion etching, etc.) and cleaning processes.

(5) During processing, a substrate is arranged on a substrate support such as a pedestal in a processing chamber of the substrate processing system. During deposition, gas mixtures including one or more precursors are introduced into the processing chamber, and plasma is struck to activate chemical reactions. A computer-controlled robot typically transfers substrates from one processing chamber to another in a sequence in which the substrates are to be processed.

SUMMARY

(6) A ring shaped structure to surround a semiconductor substrate in a processing chamber comprises an inner portion of the ring shaped structure having an inner diameter and an outer portion of the ring shaped structure having an outer diameter. The inner portion includes a first portion that descends from a top surface of the ring shaped structure, a second portion that extends from a bottom end of the first portion horizontally towards the semiconductor substrate, a third portion that descends vertically from a distal end of the second portion, a fourth portion that extends horizontally from a bottom end of the third portion towards the outer diameter, a fifth portion that descends from a distal end of the fourth portion towards the outer diameter at an acute angle relative to the second portion, and a sixth portion that extends horizontally from a bottom end of the fifth portion towards the outer portion.

(7) In other features, the top surface of the ring shaped structure is coplanar with a top surface of the semiconductor substrate, and the first portion descends vertically from the top surface of the ring shaped structure for a distance that is about a thickness of the semiconductor substrate.

(8) In another feature, the first portion is spaced from an outer edge of the semiconductor substrate by a predetermined distance.

(9) In another feature, the distal end of the second portion extends under an outer edge of the

semiconductor substrate.

(10) In another feature, a first end of the third portion forms a right angle at the distal end of the second portion.

(11) In another feature, the ring shaped structure further comprises a plurality of through holes extending from the top surface of the ring shaped structure through the fourth portion of the ring shaped structure.

(12) In another feature, the through holes have a predetermined diameter and are arranged at a predetermined radial distance from the first portion of the ring shaped structure.

(13) In another feature, the through holes descend from the top surface of the ring shaped structure to the fourth portion of the ring shaped structure at an angle other than 90 degrees.

(14) In another feature, the through holes descend vertically from the top surface of the ring shaped structure to the fourth portion of the ring shaped structure.

(15) In another feature, the through holes descend from the top surface of the ring shaped structure to the fourth portion of the ring shaped structure at a 45 degree angle relative to the first portion that descends vertically from the top surface of the ring shaped structure.

(16) In another feature, the ring shaped structure further comprises a plurality of tabs to support the semiconductor substrate.

(17) In still other features, a ring shaped structure to surround a semiconductor substrate in a processing chamber comprises an inner portion of the ring shaped structure having an inner diameter and an outer portion of the ring shaped structure having an outer diameter. The inner portion includes a first portion that descends from a top surface of the ring shaped structure, a second portion that extends from a bottom end of the first portion horizontally towards the semiconductor substrate, a third portion that descends vertically from a distal end of the second portion, a fourth portion that descends from a bottom end of the third portion towards the outer portion at a first acute angle relative to the second portion, a fifth portion that extends horizontally from a bottom end of the fourth portion towards the outer diameter, a sixth portion that descends from a distal end of the fifth portion towards the outer diameter at a second acute angle relative to the second portion, and a seventh portion that extends horizontally from a bottom end of the sixth portion towards the outer portion.

(18) In other features, the top surface of the ring shaped structure is coplanar with a top surface of the semiconductor substrate, and the first portion descends vertically from the top surface of the ring shaped structure for a distance that is about a thickness of the semiconductor substrate.

(19) In another feature, the first portion is spaced from an outer edge of the semiconductor substrate by a predetermined distance.

(20) In another feature, the distal end of the second portion extends under an outer edge of the semiconductor substrate.

(21) In another feature, a first end of the third portion forms a right angle at the distal end of the second portion.

(22) In another feature, the fourth portion descends at a 30 degree angle relative to the second portion.

(23) In another feature, the ring shaped structure further comprises a plurality of through holes extending from the top surface of the ring shaped structure through the fifth portion of the ring shaped structure.

(24) In another feature, the through holes have a predetermined diameter and are arranged at a predetermined radial distance from the first portion of the ring shaped structure.

(25) In another feature, the through holes descend from the top surface of the ring shaped structure to the fifth portion of the ring shaped structure at an angle other than 90 degrees.

(26) In another feature, the through holes descend vertically from the top surface of the ring shaped structure to the fourth portion of the ring shaped structure.

(27) In another feature, the through holes descend from the top surface of the ring shaped structure

to the fifth portion of the ring shaped structure at a 45 degree angle relative to the first portion that descends vertically from the top surface of the ring shaped structure.

(28) In another feature, the ring shaped structure further comprises a plurality of tabs to support the semiconductor substrate.

(29) In still other features, a ring shaped structure to surround a semiconductor substrate in a processing chamber comprises an inner portion of the ring shaped structure having an inner diameter and an outer portion of the ring shaped structure having an outer diameter. The inner portion includes a first portion that descends from a top surface of the ring shaped structure, a second portion that extends initially upwards and subsequently horizontally outwards towards the semiconductor substrate from a bottom end of the first portion, a third portion that descends vertically from a distal end of the second portion, a fourth portion that descends from a bottom end of the third portion towards the outer portion at a first acute angle relative to the top surface of the ring shaped structure, a fifth portion that extends horizontally from a bottom end of the fourth portion towards the outer diameter, a sixth portion that descends from a distal end of the fifth portion towards the outer diameter at a second acute angle relative to the top surface of the ring shaped structure, and a seventh portion that extends horizontally from a bottom end of the sixth portion towards the outer portion.

(30) In other features, the top surface of the ring shaped structure is coplanar with a top surface of the semiconductor substrate, and the first portion descends vertically from the top surface of the ring shaped structure for a distance greater than a thickness of the semiconductor substrate.

(31) In another feature, the first portion is spaced from an outer edge of the semiconductor substrate by a predetermined distance.

(32) In other features, a horizontal portion of the second portion is vertically spaced from a bottom surface of the semiconductor substrate by a first predetermined distance, and the third portion is horizontally spaced from an outer edge of the semiconductor substrate by a second predetermined distance.

(33) In another feature, the ring shaped structure further comprises a plurality of through holes extending from the top surface of the ring shaped structure through the fifth portion of the ring shaped structure.

(34) In another feature, the through holes have a predetermined diameter and are arranged at a predetermined radial distance from the first portion of the ring shaped structure.

(35) In another feature, the through holes descend from the top surface of the ring shaped structure to the fifth portion of the ring shaped structure at an angle other than 90 degrees.

(36) In another feature, the through holes descend vertically from the top surface of the ring shaped structure to the fourth portion of the ring shaped structure.

(37) In another feature, the through holes descend from the top surface of the ring shaped structure to the fifth portion of the ring shaped structure at a 45 degree angle relative to the first portion that descends vertically from the top surface of the ring shaped structure.

(38) In another feature, the ring shaped structure further comprises a plurality of tabs to support the semiconductor substrate.

(39) In still other features, a ring shaped structure to surround a semiconductor substrate in a processing chamber comprises an inner portion of the ring shaped structure having an inner diameter, and an outer portion of the ring shaped structure having an outer diameter. The inner portion includes a first portion that descends from a top surface of the ring shaped structure, a second portion that extends from a bottom end of the first portion horizontally towards the outer diameter of the ring shaped structure, a third portion that descends from a distal end of the second portion towards the outer diameter at an acute angle relative to the second portion, and a fourth portion that extends horizontally from a bottom end of the third portion towards the outer portion.

(40) In other features, the top surface of the ring shaped structure is coplanar with a top surface of the semiconductor substrate, and the first portion descends vertically from the top surface of the

ring shaped structure for a distance that is about a thickness of the semiconductor substrate.

(41) In another feature, the first portion is spaced from an outer edge of the semiconductor substrate by a predetermined distance.

(42) In another feature, the ring shaped structure further comprises a plurality of through holes extending from the top surface of the ring shaped structure through the second portion of the ring shaped structure.

(43) In another feature, the through holes have a predetermined diameter and are arranged at a predetermined radial distance from the first portion of the ring shaped structure.

(44) In another feature, the through holes are closer to the distal end of the second portion than to the first portion of the ring shaped structure.

(45) In another feature, the through holes descend vertically from the top surface of the ring shaped structure to the second portion of the ring shaped structure.

(46) In another feature, the ring shaped structure further comprises a plurality of tabs to support the semiconductor substrate.

(47) Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. 1 shows an example of a substrate processing system comprising a processing chamber;

(3) FIG. 2 shows a portion of a semiconductor wafer indicating areas where deposition is undesirable and where deposition is prevented or minimized by the carrier rings of the present disclosure;

(4) FIGS. 3A-3D show examples of carrier rings with and without through holes according to the present disclosure;

(5) FIGS. 4A-4C show examples of a first carrier ring design with and without through holes according to the present disclosure;

(6) FIGS. 5A-5C show examples of a second carrier ring design with and without through holes according to the present disclosure;

(7) FIGS. 6A-6C show examples of a third carrier ring design with and without through holes according to the present disclosure; and

(8) FIGS. 7A and 7B show examples of a fourth carrier ring design with and without through holes according to the present disclosure.

(9) In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

(10) During a deposition process, a material intended to be deposited only on a bottom surface of a semiconductor wafer (hereinafter “the wafer”) sometimes also gets deposited at the wafer's front side, edge, and bevel. Current carrier ring designs allow an unacceptable amount of deposition on the wafer's front side and bevel edge. In some carrier ring designs, deposition on the wafer's back side (i.e., under side or the surface opposite to the top surface) is depleted in a uniform manner at the carrier ring transition.

(11) The present disclosure provides various carrier ring designs and configurations to control the amount of deposition at the wafer's front side and bevel edge. The carrier ring designs can control the amount of deposition at all three locations of the wafer: front side of bevel, edge and back side of bevel, and front and back side of the wafer (with no deposition desired on the front side).

Specifically, edge profiles of the carrier rings are designed to control flow of process gases, flow of front side purge gas, and plasma effects. In some designs, through holes are added to the carrier rings to control gas flows. The edge profiles and added features can reduce or eliminate deposition at the wafer's front side and bevel edge. These and other features of the carrier ring designs and configurations according to the present disclosure are now described below in detail.

(12) The present disclosure is organized as follows. Initially, an example of a processing chamber is shown and described with reference to FIG. 1. Thereafter, the problem being solved by the carrier ring designs of the present disclosure is shown and described with reference to FIG. 2.

Subsequently, various designs and structural details of the carrier rings according to the present disclosure are shown and described with reference to FIGS. 3A-7B.

(13) FIG. 1 shows an example of a substrate processing system **10** including a processing chamber **12** that encloses other components of the substrate processing system **10** and contains RF plasma (if used). The processing chamber **12** includes a showerhead **14** and a substrate support assembly **16**. A substrate **18** is arranged on the substrate support assembly **16**. As explained below, the carrier rings of the present disclosure are used with the substrate support assembly **16** and include a plurality of tabs that support the substrate **18** during processing. Process gases and vaporized chemistry are introduced through the substrate support assembly **16** for depositing material on the back (bottom) side of the substrate **18**. The showerhead **14** introduces a purge gas to prevent deposition on the front (top) side of the substrate **18**.

(14) If plasma is used, the plasma can be direct or remote plasma. In this example, an RF generating system **30** generates and outputs an RF voltage to the showerhead **14**. The substrate support assembly **16** is grounded (the other is DC grounded, AC grounded, or floating). For example only, the RF generating system **30** may include an RF voltage generator **32** that generates the RF voltage that is fed by a matching network **34** to the showerhead **14**. Alternately, the plasma may be delivered by a remote plasma source (not shown).

(15) A gas delivery system **40** includes one or more gas sources **42-1**, . . . , **42-(N-1)**, and **42-N** (collectively gas sources **42**), where N is a positive integer. The gas sources **42** supply one or more process gases, precursor gases, cleaning gases, purge gases, etc. to the processing chamber **12**. The gas sources **42** are connected by valves **44-1**, . . . , **44-(N-1)**, and **44-N** (collectively valves **44**) and mass flow controllers **46-1**, . . . , **46-(N-1)**, and **46-N** (collectively mass flow controllers **46**) to a manifold **48**. An output of the manifold **48** is fed to the substrate support assembly **16**. Vaporized precursors may also be used. For some vaporized precursors, no MFC is used. The gas delivery system **40** delivers a purge gas to the showerhead **14**.

(16) A heater controller **50** may be connected to a heater element (not shown) arranged in the substrate support assembly **16** and in the showerhead **14**. The heater controller **50** may be used to control a temperature of the showerhead **14**, the substrate support assembly **16** and the substrate **18**. A valve **60** and pump **62** may be used to evacuate reactants from the processing chamber **12**. A controller **70** may be used to control components of the substrate processing system **10**. For example only, the controller **70** may be used for controlling flow of process gases and purge gases, monitoring process parameters such as temperature, pressure, power, etc., striking and extinguishing plasma, removing reactants, etc.

(17) FIG. 2 shows a portion of a wafer with its bevel edge. As indicated, deposition on the front side and bevel edge of the wafer is undesirable. The carrier ring designs according to the present disclosure can prevent or minimize the deposition on the front side and bevel edge of the wafer (specifically, on the portion of the wafer's bevel edge that is above the horizontal dashed line).

(18) Below are various designs of carrier rings that are arranged around a wafer in a processing chamber. Views of complete rings are shown only for the first carrier ring design. Views of complete rings are omitted for the second, third, and fourth carrier designs. However, the geometric and structural details are shown and described for all four carrier ring designs.

(19) FIGS. 3A-3D show examples of a carrier ring according to the first design of the present

disclosure. FIGS. 3A and 3B show a carrier ring **200** without through holes according to the first design. FIG. 3A shows a top view of the carrier ring **200** without through holes according to the first design. FIG. 3B shows a bottom view of the carrier ring **200** without through holes according to the first design.

(20) As shown, the carrier ring **200** according to the first design (and all other carrier ring designs with and without through holes described below) is annular, has an inner diameter (ID) and an outer diameter (OD). The inner diameter (ID) generally surrounds a wafer (element **202** shown in FIGS. 4A-7B) in a processing chamber. The carrier ring **200** according to the first design (and all other carrier ring designs with and without through holes described below) includes a plurality of tabs **201** (e.g., six tabs are shown in the example) to support the wafer **202** during processing. The top surface of the carrier ring **200** is generally coplanar with the top surface of the wafer **202**. The geometry and structural profile or shape of the inner edge of the carrier ring **200** indicated by a dotted oval is shown and described below in detail with reference to FIGS. 4A-4C. The geometry and structural profile or shape of the inner edge (i.e., the design of surface profile) of each of the four carrier ring designs is different as described below with reference to FIGS. 4A-7B in detail.

(21) FIGS. 3C and 3D show examples of a carrier ring **250** with through holes according to the first design. The only difference between the carrier rings **200** and **250** is that the carrier ring **200** is without the through holes, and the carrier ring **250** is with the through holes. The carrier rings **200** and **250** are identical in all other respects. FIG. 3C shows a top view of the carrier ring **250** with through holes according to the first design. FIG. 3D shows a bottom view of the carrier ring **250** with through holes according to the first design. The geometry and structural profile or shape of the inner edge of the carrier ring **250** indicated by a dotted oval is shown and described below in detail with reference to FIGS. 4A-4C. As described below in detail, the geometry and arrangement of the through holes differ for each of the four carrier ring designs.

(22) FIGS. 4A-4C show the first design (i.e., the first design of surface profile) of the carrier rings **200** and **250** according to the present disclosure in detail. The carrier rings **200** and **250** are annular or have a ring shaped structure to surround a semiconductor substrate (e.g., the wafer **202**) in a processing chamber. FIG. 4A shows the carrier ring **200** without through holes. FIG. 4B shows the carrier ring **250** with through holes. FIG. 4C shows the structural details of the inner portion of the carrier rings **200** and **250** that are identical. The structural details of the carrier rings **200** and **250** are described below with reference to the carrier ring **250**. It should be understood that other than the through holes, the description of the carrier ring **250** is identical for the carrier ring **200**.

(23) The carrier ring **250** comprises an inner portion having an inner diameter (ID) and an outer portion having an outer diameter (OD). The inner portion of the carrier ring **250** includes a first portion **252** that descends from a top surface **254** of the carrier ring **250**. A second portion **256** of the carrier ring **250** extends from a bottom end of the first portion **252** of the carrier ring **250** horizontally (i.e., parallel to the top surface **254** of the carrier ring **250**) towards the wafer **202**. A third portion **258** of the carrier ring **250** descends vertically (i.e., perpendicular to the top surface **254** of the carrier ring **250**) from a distal end of the second portion **256** of the carrier ring **250**. A fourth portion **260** of the carrier ring **250** extends horizontally (i.e., parallel to the top surface **254** of the carrier ring **250**) from a bottom end of the third portion **258** of the carrier ring **250** towards the outer diameter (OD) of the carrier ring **250**.

(24) A fifth portion **262** of the carrier ring **250** descends from a distal end of the fourth portion **260** of the carrier ring **250** towards the outer diameter (OD) or the outer portion of the carrier ring **250** at an acute angle relative to the second portion **256** and the top surface **254** of the carrier ring **250**. A sixth portion **264** of the carrier ring **250** extends horizontally (i.e., parallel to the top surface **254** of the carrier ring **250**) from a bottom end of the fifth portion **262** of the carrier ring **250** towards the outer portion or the outer diameter (OD) of the carrier ring **250**.

(25) The top surface **254** of the carrier ring **250** is coplanar with a top surface of the wafer **202**. The first portion **252** of the carrier ring **250** descends vertically (i.e., perpendicularly) from the top

surface **254** of the carrier ring **250** for a distance that is approximately equal to a thickness of the wafer **202**. The first portion **252** of the carrier ring **250** is spaced from an outer edge (e.g., an OD) of the wafer **202** by a predetermined distance. The distal end of the second portion **256** of the carrier ring **250** extends under the outer edge of the wafer **202**. A first end of the third portion **258** of the carrier ring **250** forms a right angle (i.e., a sharp corner) at the distal end of the second portion **256** of the carrier ring **250**.

(26) The carrier ring **250** (but not the carrier ring **200**) comprises a plurality of through holes **270** extending from the top surface **254** of the carrier ring **250** through the fourth portion **260** of the carrier ring **250**. The through holes **270** of the carrier ring **250** have a predetermined diameter and are arranged at a predetermined radial distance from the first portion **252** of the carrier ring **250**. The through holes **270** of the carrier ring **250** descend from the top surface **254** of the carrier ring **250** to the fourth portion **260** of the carrier ring **250** at an angle other than 90 degrees. For example, the through holes **270** of the carrier ring **250** descend from the top surface **254** of the carrier ring **250** to the fourth portion **260** of the carrier ring **250** at a 45 degree angle relative to the first portion **252** that descends vertically (i.e., perpendicularly) from the top surface **254** of the carrier ring **250**. Stated differently, the through holes **270** of the carrier ring **250** descend from the top surface **254** of the carrier ring **250** to the fourth portion **260** of the carrier ring **250** at a 45 degree angle relative to the top surface **254** of the carrier ring **250**.

(27) FIGS. 5A-5C show the second design (i.e., the second design of surface profile) of carrier rings **300** and **350** according to the present disclosure in detail. The carrier rings **300** and **350** are annular or have a ring shaped structure to surround a semiconductor substrate (e.g., the wafer **202**) in a processing chamber. FIG. 5A shows the carrier ring **300** without through holes. FIG. 5B shows the carrier ring **350** with through holes. FIG. 5C shows the structural details of the inner portion of the carrier rings **300** and **350** that are identical. The structural details of the carrier rings **300** and **350** are described below with reference to the carrier ring **350**. It should be understood that other than the through holes, the description of the carrier ring **350** is identical for the carrier ring **300**.

(28) The carrier ring **350** comprises an inner portion having an inner diameter (ID) and an outer portion having an outer diameter (OD). The inner portion of the carrier ring **350** includes a first portion **352** that descends from a top surface **354** of the carrier ring **350**. A second portion **356** of the carrier ring **350** extends from a bottom end of the first portion **352** horizontally (i.e., parallel to the top surface **354** of the carrier ring **350**) towards the wafer **202**. A third portion **358** of the carrier ring **350** descends vertically (i.e., perpendicular to the top surface **354** of the carrier ring **350**) from a distal end of the second portion **356** of the carrier ring **350**. A fourth portion **360** of the carrier ring **350** descends from a bottom end of the third portion **358** of the carrier ring **350** towards the outer diameter (OD) or the outer portion of the carrier ring **350** at a first acute angle relative to the second portion **356** or the top surface **354** of the carrier ring **350**.

(29) A fifth portion **362** of the carrier ring **350** extends horizontally (i.e., parallel to the top surface **354** of the carrier ring **350**) from a bottom end of the fourth portion **360** of the carrier ring **350** towards the outer diameter (OD) or the outer portion of the carrier ring **350**. A sixth portion **364** of the carrier ring **350** descends from a distal end of the fifth portion **362** of the carrier ring **350** towards the outer diameter (OD) or the outer portion of the carrier ring **350** at a second acute angle relative to the second portion **356** or the top surface **354** of the carrier ring **350**. A seventh portion **366** of the carrier ring **350** extends horizontally (i.e., parallel to the top surface **354** of the carrier ring **350**) from a bottom end of the sixth portion **364** of the carrier ring **350** towards the outer portion or the outer diameter (OD) of the carrier ring **350**.

(30) The top surface **354** of the carrier ring **350** is coplanar with a top surface of the wafer **202**. The first portion **352** of the carrier ring **350** descends vertically (i.e., perpendicularly) from the top surface **354** of the carrier ring **350** for a distance that is approximately equal to a thickness of the wafer **202**. The first portion **352** of the carrier ring **350** is spaced from an outer edge (e.g., an OD) of the wafer **202** by a predetermined distance. The distal end of the second portion **356** of the

carrier ring 350 extends under the outer edge of the wafer 202. A first end of the third portion 358 of the carrier ring 350 forms a right angle (i.e., a sharp corner) at the distal end of the second portion 356 of the carrier ring 350. The fourth portion 360 of the carrier ring 350 descends at a 30 degree angle relative to the second portion 356 and the top surface 354 of the carrier ring 350.

(31) The carrier ring 350 (but not the carrier ring 300) comprises a plurality of through holes 370 extending from the top surface 354 of the carrier ring 350 through the fifth portion 362 of the carrier ring 350. The through holes 370 of the carrier ring 350 have a predetermined diameter and are arranged at a predetermined radial distance from the first portion 352 of the carrier ring 350. The through holes 370 of the carrier ring 350 descend from the top surface 354 of the carrier ring 350 to the fifth portion 362 of the carrier ring 350 at an angle other than 90 degrees. For example, the through holes 370 descend from the top surface 354 of the carrier ring 350 to the fifth portion 362 of the carrier ring 350 at a 45 degree angle relative to the first portion 352 of the carrier ring 350 that descends vertically (i.e., perpendicularly) from the top surface 354 of the carrier ring 350. Stated differently, the through holes 370 of the carrier ring 350 descend from the top surface 354 of the carrier ring 350 to the fifth portion 362 of the carrier ring 350 at a 45 degree angle relative to the top surface 354 of the carrier ring 350.

(32) FIGS. 6A-6C show the third design (i.e., the third design of surface profile) of carrier rings 400 and 450 according to the present disclosure in detail. The carrier rings 400 and 450 are annular or have a ring shaped structure to surround a semiconductor substrate (e.g., the wafer 202) in a processing chamber. FIG. 6A shows the carrier ring 400 without through holes. FIG. 6B shows the carrier ring 450 with through holes. FIG. 6C shows the structural details of the inner portion of the carrier rings 400 and 450 that are identical. The structural details of the carrier rings 400 and 450 are described below with reference to the carrier ring 450. It should be understood that other than the through holes, the description of the carrier ring 450 is identical for the carrier ring 400.

(33) The carrier ring 450 comprises an inner portion having an inner diameter (ID) and an outer portion having an outer diameter (OD). The inner portion of the carrier ring 450 includes a first portion 452 that descends from a top surface 454 of the carrier ring 450. From a bottom end of the first portion 452 of the carrier ring 450, a second portion 456 of the carrier ring 450 extends initially slightly upwards and subsequently horizontally (i.e., parallel to the top surface 454 of the carrier ring 450) outwards towards the wafer 202. A third portion 458 of the carrier ring 450 descends vertically (i.e., perpendicular to the top surface 454 of the carrier ring 450) from a distal end of the second portion 456 of the carrier ring 450. A fourth portion 460 of the carrier ring 450 descends from a bottom end of the third portion 458 of the carrier ring 450 towards the outer portion or the outer diameter (OD) of the carrier ring 450 at a first acute angle relative to the top surface 454 of the carrier ring 450.

(34) A fifth portion 462 of the carrier ring 450 extends horizontally (i.e., parallel to the top surface 454 of the carrier ring 450) from a bottom end of the fourth portion 460 of the carrier ring 450 towards the outer diameter (OD) or the outer portion of the carrier ring 450. A sixth portion 464 of the carrier ring 450 descends from a distal end of the fifth portion 462 of the carrier ring 450 towards the outer diameter (OD) or the outer portion of the carrier ring 450 at a second acute angle relative to the top surface 454 of the carrier ring 450. A seventh portion 466 of the carrier ring 450 extends horizontally (i.e., parallel to the top surface 454 of the carrier ring 450) from a bottom end of the sixth portion 464 of the carrier ring 450 towards the outer portion or the outer diameter (OD) of the carrier ring 450.

(35) The top surface 454 of the carrier ring 450 is coplanar with a top surface of the wafer 202. The first portion 452 descends vertically (i.e., perpendicularly) from the top surface 454 of the carrier ring 450 for a distance greater than a thickness of the wafer 202. The first portion 452 of the carrier ring 450 is spaced from an outer edge (e.g., an OD) of the wafer 202 by a predetermined distance. The horizontal portion of the second portion 456 of the carrier ring 450 is vertically spaced from a bottom surface of the wafer 202 by a first predetermined distance. The third portion 458 of the

carrier ring **450** is horizontally spaced from an outer edge (e.g., the OD) of the wafer **202** by a second predetermined distance. The first and second predetermined distances may be equal.

(36) The carrier ring **450** (but not the carrier ring **400**) further comprises a plurality of through holes **470** extending from the top surface **454** of the carrier ring **450** through the fifth portion **462** of the carrier ring **450**. The through holes **470** of the carrier ring **450** have a predetermined diameter and are arranged at a predetermined radial distance from the first portion **452** of the carrier ring **450**. The through holes **470** of the carrier ring **450** descend from the top surface **454** of the carrier ring **450** to the fifth portion **462** of the carrier ring **450** at an angle other than 90 degrees. For example, the through holes **470** of the carrier ring **450** descend from the top surface **454** of the carrier ring **450** to the fifth portion **462** of the carrier ring **450** at a 45 degree angle relative to the first portion **452** that descends vertically (i.e., perpendicularly) from the top surface **454** of the carrier ring **450**. Stated differently, the through holes **470** of the carrier ring **450** descend from the top surface **454** of the carrier ring **450** to the fifth portion **462** of the carrier ring **450** at a 45 degree angle relative to the top surface **454** of the carrier ring **450**.

(37) FIGS. 7A and 7B show the fourth design (i.e., the fourth design of surface profile) of carrier rings **500** and **550** according to the present disclosure in detail. The carrier rings **500** and **550** are annular or have a ring shaped structure to surround a semiconductor substrate (e.g., the wafer **202**) in a processing chamber. FIG. 7A shows the carrier ring **500** without through holes. FIG. 7B shows the carrier ring **550** with through holes. The structural details of the carrier rings **500** and **550** are identical except that the carrier ring **500** is without through holes and the carrier ring **550** is with the through holes. Accordingly, the structural details of the carrier rings **500** and **550** are described below with reference to the carrier ring **550**. It should be understood that other than the through holes, the description of the carrier ring **550** is identical for the carrier ring **500**.

(38) The carrier ring **550** comprises an inner portion having an inner diameter (ID) and an outer portion having an outer diameter (OD). The inner portion of the carrier ring **550** includes a first portion **552** that descends from a top surface **554** of the carrier ring **550**. A second portion **556** of the carrier ring **550** extends from a bottom end of the first portion **552** of the carrier ring **550** horizontally (i.e., parallel to the top surface **554** of the carrier ring **550**) towards the outer diameter (OD) or the outer portion of the carrier ring **550**. A third portion **558** of the carrier ring **550** descends from a distal end of the second portion **556** of the carrier ring **550** towards the outer diameter (OD) or the outer portion of the carrier ring **550** at an acute angle relative to the second portion **556** or the top surface **554** of the carrier ring **550**. A fourth portion **560** of the carrier ring **550** extends horizontally (i.e., parallel to the top surface **554** of the carrier ring **550**) from a bottom end of the third portion **558** of the carrier ring **550** towards the outer portion or the outer diameter (OD) of the carrier ring **550**.

(39) The top surface **554** of the carrier ring **550** is coplanar with a top surface of the wafer **202**. The first portion **552** of the carrier ring **550** descends vertically (i.e., perpendicularly) from the top surface **554** of the carrier ring **550** for a distance that is approximately equal to a thickness of the wafer **202**. The first portion **552** of the carrier ring **550** is spaced from an outer edge (e.g., an OD) of the wafer **202** by a predetermined distance.

(40) The carrier ring **550** further comprises a plurality of through holes **570** extending from the top surface **454** of the carrier ring **550** through the second portion **556** of the carrier ring **550**. The through holes **570** of the carrier ring **550** have a predetermined diameter and are arranged at a predetermined radial distance from the first portion **552** of the carrier ring **550**. The through holes **570** of the carrier ring **550** are closer to the distal end of the second portion **556** of the carrier ring **550** than to the first portion **552** of the carrier ring **550**. The through holes **570** of the carrier ring **550** descend vertically (i.e., perpendicularly) from the top surface **554** of the carrier ring **550** to the second portion **556** of the carrier ring **550**.

(41) The following description is common for the above carrier ring designs. All the junctions between the various portions (e.g., between the top surface and the first portion, the first portion

and the second portion, and so on) of the inner portions of the carrier rings are rounded or curved, except where a junction between two portions is specifically described as being or forming a sharp corner.

(42) The inner diameter (ID) of the carrier ring determines a gap between the wafer edge and the carrier ring ID. The smaller the gap, the greater the reduction in the concentration of the deposition material at the wafer's front side. However, changing (i.e., increasing or decreasing) the carrier ring ID beyond an optimal ID has an effect on the concentration of the deposition material at the wafer's front side. For example, increasing the ID beyond the optimal ID increases deposition on the wafer's front side and bevel/edge. Decreasing the ID further beyond the optimal ID can increase the concentration of deposition material at the wafer's front side.

(43) The profile angle (i.e., acute angle of a portion) of the inner portion of the carrier rings shown in FIGS. 5A-6C can be between 15 and 45 degrees, preferably 30 degrees. Changing the lower profile of the inner portion of the carrier rings can have some effect on the concentration of deposition material at the wafer's back side and bevel edge. The 30 degree angle affects deposition on the whole back side of the wafer (i.e., affects the uniformity of film deposition).

(44) The through holes are radially located on the carrier ring. The through holes are equidistant from each other. The total number of the through holes can be about 120. The radial location of the through holes is such that the through holes are closer to the ID than the OD of the carrier rings. Moving the through holes towards the OD of the carrier rings only minimally increases the concentration of deposition material at the wafer's front side and bevel/edge with no effect to the deposition of material at the wafer's backside.

(45) The diameter of the through holes can be about 1 mm or smaller or larger. A smaller diameter can decrease the concentration of the deposition material at the wafer's front side and bevel/edge with no effect to the deposition of material at the wafer's backside. Making the through holes larger beyond an optimum diameter increases front side and bevel/edge deposition. The diameter of the through holes has a larger effect than the radial location of the through holes on the concentration of deposition material at the wafer's bevel/edge and backside.

(46) Adding the through holes provides better control in preventing deposition on the wafer's bevel/edge and front side without any detrimental effect on film uniformity on the wafer's back side. The through holes can be vertical or slanted relative to the top surface of the carrier ring. The angle of the through holes can be between 30 degrees and 90 degrees (i.e., vertical) relative to the top surface of the carrier ring. An angle of 45 degrees is preferable to minimize purge gas flow losses as well as to minimize deposition on the wafer's bevel/edge and front side. An angle of 90 degrees is easier to machine (i.e., manufacture).

(47) The foregoing description is merely illustrative in nature and is not intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure. Further, although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

(48) Spatial and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms, including "connected," "engaged," "coupled," "adjacent," "next to," "on top of," "above," "below," and "disposed." Unless explicitly described as being "direct," when a relationship between first and second

elements is described in the above disclosure, that relationship can be a direct relationship where no other intervening elements are present between the first and second elements, but can also be an indirect relationship where one or more intervening elements are present (either spatially or functionally) between the first and second elements. As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

(49) In some implementations, a controller is part of a system, which may be part of the above-described examples. Such systems can comprise semiconductor processing equipment, including a processing tool or tools, chamber or chambers, a platform or platforms for processing, and/or specific processing components (a wafer pedestal, a gas flow system, etc.). These systems may be integrated with electronics for controlling their operation before, during, and after processing of a semiconductor wafer or substrate. The electronics may be referred to as the “controller,” which may control various components or subparts of the system or systems. The controller, depending on the processing requirements and/or the type of system, may be programmed to control any of the processes disclosed herein, including the delivery of processing gases, temperature settings (e.g., heating and/or cooling), pressure settings, vacuum settings, power settings, radio frequency (RF) generator settings, RF matching circuit settings, frequency settings, flow rate settings, fluid delivery settings, positional and operation settings, wafer transfers into and out of a tool and other transfer tools and/or load locks connected to or interfaced with a specific system.

(50) Broadly speaking, the controller may be defined as electronics having various integrated circuits, logic, memory, and/or software that receive instructions, issue instructions, control operation, enable cleaning operations, enable endpoint measurements, and the like. The integrated circuits may include chips in the form of firmware that store program instructions, digital signal processors (DSPs), chips defined as application specific integrated circuits (ASICs), and/or one or more microprocessors, or microcontrollers that execute program instructions (e.g., software). Program instructions may be instructions communicated to the controller in the form of various individual settings (or program files), defining operational parameters for carrying out a particular process on or for a semiconductor wafer or to a system. The operational parameters may, in some embodiments, be part of a recipe defined by process engineers to accomplish one or more processing steps during the fabrication of one or more layers, materials, metals, oxides, silicon, silicon dioxide, surfaces, circuits, and/or dies of a wafer.

(51) The controller, in some implementations, may be a part of or coupled to a computer that is integrated with the system, coupled to the system, otherwise networked to the system, or a combination thereof. For example, the controller may be in the “cloud” or all or a part of a fab host computer system, which can allow for remote access of the wafer processing. The computer may enable remote access to the system to monitor current progress of fabrication operations, examine a history of past fabrication operations, examine trends or performance metrics from a plurality of fabrication operations, to change parameters of current processing, to set processing steps to follow a current processing, or to start a new process.

(52) In some examples, a remote computer (e.g. a server) can provide process recipes to a system over a network, which may include a local network or the Internet. The remote computer may include a user interface that enables entry or programming of parameters and/or settings, which are then communicated to the system from the remote computer. In some examples, the controller receives instructions in the form of data, which specify parameters for each of the processing steps to be performed during one or more operations. It should be understood that the parameters may be specific to the type of process to be performed and the type of tool that the controller is configured to interface with or control.

(53) Thus as described above, the controller may be distributed, such as by comprising one or more discrete controllers that are networked together and working towards a common purpose, such as the processes and controls described herein. An example of a distributed controller for such

purposes would be one or more integrated circuits on a chamber in communication with one or more integrated circuits located remotely (such as at the platform level or as part of a remote computer) that combine to control a process on the chamber.

(54) Without limitation, example systems may include a plasma etch chamber or module, a deposition chamber or module, a spin-rinse chamber or module, a metal plating chamber or module, a clean chamber or module, a bevel edge etch chamber or module, a physical vapor deposition (PVD) chamber or module, a chemical vapor deposition (CVD) chamber or module, an atomic layer deposition (ALD) chamber or module, an atomic layer etch (ALE) chamber or module, an ion implantation chamber or module, a track chamber or module, and any other semiconductor processing systems that may be associated or used in the fabrication and/or manufacturing of semiconductor wafers.

(55) As noted above, depending on the process step or steps to be performed by the tool, the controller might communicate with one or more of other tool circuits or modules, other tool components, cluster tools, other tool interfaces, adjacent tools, neighboring tools, tools located throughout a factory, a main computer, another controller, or tools used in material transport that bring containers of wafers to and from tool locations and/or load ports in a semiconductor manufacturing factory.

Claims

1. A carrier ring for supporting a substrate during semiconductor processing operations, the carrier ring comprising a ring structure having an inner diameter and an outer diameter, wherein: the ring structure has a cross-sectional profile having at least a first portion, a second portion, a third portion, a fourth portion, a fifth portion, and a sixth portion; the ring structure has a top surface that connects with the first portion; the first portion connects with the second portion and extends along a direction perpendicular to the top surface; the second portion connects with the third portion; the third portion connects with the fourth portion and extends along a direction perpendicular to the top surface; the fifth portion connects with the fourth portion and is parallel to the top surface; the sixth portion connects with the fifth portion; the first portion is located between parallel planes defined by the top surface and the second portion; the third portion is located radially inward of the first portion; at least part of the fourth portion is linear and at a first acute angle relative to the top surface; and the sixth portion is at a second acute angle relative to the top surface.
2. The carrier ring of claim 1, further comprising a seventh portion that connects with the sixth portion, wherein the seventh portion is parallel to the top surface.
3. The carrier ring of claim 2, wherein the first acute angle is between 15° and 45°.
4. The carrier ring of claim 3, wherein the first acute angle is 30°.
5. The carrier ring of claim 4, further comprising a circular array of holes extending around the ring structure, each hole extending through the ring structure from the top surface.
6. The carrier ring of claim 5, wherein: each hole has a first end that exits the ring structure at the top surface and a second end opposite the first end, and the second ends of the holes are positioned radially outward of the first ends of the holes.
7. The carrier ring of claim 6, wherein the second ends of the holes exit the ring structure in the fifth portion.
8. The carrier ring of claim 6, wherein each hole extends along an axis that is at an angle of between 30° and 90° relative to the top surface.
9. The carrier ring of claim 7, wherein each hole extends along an axis that is at a 45° angle relative to the top surface.
10. The carrier ring of claim 2, wherein the connection between the fourth portion and the fifth portion is rounded.
11. The carrier ring of claim 10, wherein the connection between the fifth portion and the sixth

portion is rounded.

12. The carrier ring of claim 11, wherein the connection between the sixth portion and the seventh portion is rounded.

13. The carrier ring of claim 12, wherein the connection between the third portion and the fourth portion is rounded.

14. The carrier ring of claim 13, wherein the connection between the first portion and the second portion is rounded.

15. The carrier ring of claim 14, wherein the connection between the first portion and the top surface is rounded.

16. The carrier ring of claim 15, further comprising a plurality of protrusions protruding from the seventh portion.
